Bigollφ et al.

A generative template for physical observables

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Abstract

We describe a method for extracting physical observables—from nuclear radii to stellar temperatures to semiconductor band gaps—using a single generative template derived from the critical Fibonacci Hamiltonian. The method requires one axiom ($\varphi^2 = \varphi + 1$, where φ is the golden ratio), one existence condition ($V = 2J$, the Aubry–André self-dual critical point), and zero adjustable parameters. The output is a set of twelve dimensionless primitives and five Cantor-sector positions, which combine multiplicatively with an appropriate physical scale to predict any observable at any bracket (energy/length scale). We validate the method against 100+ independently measured quantities across 15 physical domains, finding sub-1% agreement in the majority of cases. The central result is the W theorem: $W \times \varphi^4 = 2 + \varphi^{1/\varphi^2}$, an exact identity (verified to machine precision, error 2.22×10^{-16}) that connects the universal gap fraction to the axiom through the thermodynamic pressure of the trace map. We present this work with deep gratitude to the mathematicians and physicists whose decades of rigorous spectral theory made the synthesis possible.

Keywords: Fibonacci Hamiltonian, Aubry–André–Harper model, golden ratio, quasicrystal, Cantor spectrum, trace map, thermodynamic pressure, spectral dimension, Hausdorff dimension, gap labeling, Chern numbers, quantum error-correcting code, unified theory, universal formula, universal theory, theory of everything, fine structure constant, cosmological constant, dark energy, dark matter, baryon fraction, galaxy rotation curves, MOND, Hofstadter butterfly, Anderson localization, metallic means, Fibonacci quasicrystal, Penrose tiling, critical coupling, self-duality, fractal spectrum, W theorem, Cantor node architecture, bracket scale system, condensation sequence, electroweak masses, Weinberg angle, Higgs mass, semiconductor band gap, atomic radii, proton charge radius, Jacobson thermodynamic gravity, Regge calculus, Bowen equation, Ruelle operator, Lyapunov exponent, multifractal, nematic–smectic A transition, quantum Hall plateau, GABA microtubule gate, nuclear transduction, Fibonacci anyons, quasiperiodic operator, Li–Boyle QECC; Fibonacci, Aubry, André, Harper, Hofstadter, Kohmoto, Kadanoff, Tang, Sutherland, Sütő, Bellissard, Jacobson, Damanik, Gorodetski, Yessen, Embree, Tcheremchantsev, Maciá, Jagannathan, Li, Boyle, Bowen, Avila, Jitomirskaya, Penrose, de Gennes, McMillan, Wen, Vishwanath, Lapierre, Fan, Milgrom, Verlinde, Van Raamsdonk, Persic, Salucci, Hamber, Kagel, Cheeger, Müller, Schrader, Hofstadter, Regge, Husmann

Introduction

The Aubry–André–Harper (AAH) Hamiltonian describes a particle on a one-dimensional lattice with a quasiperiodic potential at irrational frequency α. When $\alpha = 1/\varphi$ (the reciprocal of the golden ratio) and the coupling is tuned to the self-dual critical point $V = 2J$, the energy spectrum becomes a Cantor set of measure zero with Hausdorff dimension approximately 0.37. This has been known since the work of Kohmoto, Kadanoff, and Tang (1983) and rigorously established by Sütő (1987, 1989), Bellissard (1992), and Damanik, Gorodetski, and Yessen (2016).

What has not been attempted—in any of the 49 papers citing DGY (2016) or the broader quasicrystal literature—is to extract fundamental coupling constants from this gap structure and test them against experiment across multiple physical domains. The present method article describes how to do exactly that. We emphasize that every formula we present is built entirely from the spectral invariants of the critical Fibonacci Hamiltonian; no quantity is fitted to data. The remarkable agreement with experiment (when it occurs) and the clear disagreements (when they occur) are reported transparently.

We publish under the collective name *Bigollo et al.* to honor Leonardo of Pisa (c. 1170–1250), known as Fibonacci or "Bigollo," whose sequence $F_n = F_{n-1} + F_{n-2}$ is the single generative object from which the entire method flows. We owe an enormous debt to the mathematicians listed in our acknowledgments; we merely connected their results in a way that, to our knowledge, had not been tried before.

Standing on the Shoulders of Giants

This framework synthesizes decades of rigorous mathematics on quasiperiodic operators, dynamical systems, spectral theory, and quantum information. The table below summarizes the indispensable contributions upon which every result in this article depends. We have added nothing to the mathematics itself—our contribution is the observation that these spectral invariants, when multiplied by appropriate physical scales, reproduce measured quantities across nuclear, atomic, stellar, galactic, and cosmological physics.

Researcher(s)	Year	Contribution	Key Result
Leonardo Fibonacci (Bigollo Pisano)	c. 1202	Liber Abaci — introduced the sequence and substitution rule	$F(n) = F(n-1) + F(n-2)$; $\varphi = \lim F(n+1)/F(n)$
Aubry & André	1980	Self-duality and metal–insulator transition at $V = 2J$	Critical coupling is the existence condition
Kohmoto, Kadanoff & Tang	1983	Trace-map renormalization for quasicrystal spectra	$x(n+1) = 2x(n)x(n-1) - x(n-2)$
Sütő	1987–89	Spectrum membership via bounded trace orbits; Lebesgue measure zero	$E \in \Sigma \Leftrightarrow$ trace orbit bounded
Bellissard	1992	Gap labeling theorem for quasiperiodic operators	IDS at gaps $\in \mathbb{Q}(\sqrt{5})$ (algebraic)
Jacobson	1995	Thermodynamic derivation of Einstein field equations	$\delta Q = TdS \rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$

Damanik, Embree, Gorodetski & Tcheremchantsev	2008	Fractal dimension scaling at large coupling	dim(Σ) · log(λ) → log(1+√2)
Avila & Jitomirskaya	2009	The Ten Martini Problem — all gaps open for all irrational α	Spectrum is a Cantor set
Damanik, Gorodetski & Yessen (DGY)	2016	The Fibonacci Hamiltonian (Inventiones Math. 206)	d(λ) = h_top/χ_u(μ_max); P(t); all gaps open
Maciá	2017	Palindromic cluster decomposition of the spectrum	Two-cluster structure → five sectors
Jagannathan	2021	Fibonacci quasicrystal review (Rev. Mod. Phys. 93)	Chern numbers: +2, −1, +1, −2; 1D → higher-D
Li & Boyle	2023	Fibonacci tiling is a quantum error-correcting code	Knill–Laflamme conditions; infinite code distance
Wen, Fan, Vishwanath & Lapierre	2021	Fibonacci-driven conformal field theories	Entanglement ~ log(t) × f(Lyapunov)
Harper	1955	Electron on 2D lattice in magnetic field	The Harper/Hofstadter equation
Hofstadter	1976	Hofstadter butterfly — fractal spectrum vs magnetic flux	Energy spectrum is a fractal of flux α
Bowen	1970s	Thermodynamic formalism for dynamical systems	Bowen's equation: P(d_H) = 0
de Gennes	1972	Liquid crystal free energy — N-SmA transition	Free energy → discretize → AAH
McMillan	1971	Smectic A ordering and the McMillan ratio	V/J = 2 at second-order N-SmA
Penrose	1974	Penrose tiling — aperiodic tiling of the plane	2D quasicrystal; 5-fold symmetry
Milgrom	1983	Modified Newtonian Dynamics (MOND)	$a_0 \approx 1.2\times10^{-10}$ m/s²
Hamber & Kagel	2004	Bianchi identity on simplicial complexes	∇μGμν = 0 exact on backbone
Cheeger, Müller & Schrader	1984	Regge calculus continuum limit	S_Regge → S_EH as mesh → 0
Verlinde	2011	Entropic gravity — gravity as emergent force	F = T∇S (gravity from information)
Van Raamsdonk	2010	Entanglement builds spacetime geometry	Less entanglement = more curvature
Persic & Salucci	1996	Universal rotation curve of spiral galaxies	Universal shape across 3 orders of mass

Method

Step 1: The Axiom and Hamiltonian

Begin with the single algebraic identity $\varphi^2 = \varphi + 1$, where $\varphi = (1 + \sqrt{5})/2 = 1.6180339887$. Construct the AAH Hamiltonian on a one-dimensional lattice of D = 233 sites (= F(13), a Fibonacci number) with on-site potential $V_n = 2\cdot\cos(2\pi n/\varphi)$ and nearest-neighbor hopping J = 1. Setting V = 2J places the system at the self-dual critical point. Diagonalize numerically (standard eigensolver; we use NumPy). The output is 233 eigenvalues forming a Cantor set with 34 gaps.

We note that D = 233 is not a free parameter. The spectral constants converge at any Fibonacci lattice size D ≥ 89; D = 233 and D = 377 differ by 0.0004% in all extracted quantities. The constants are a convergent limit of the D → ∞ spectrum, not a parameter choice.

Step 2: Extract the Twelve Primitives

From the eigenvalue spectrum, extract the following dimensionless constants. Each is a ratio of eigenvalue positions or gap widths, requiring no physical units:

#	Primitive	Value	Extraction
1	φ (golden ratio)	1.6180	Axiom
2	$H = \varphi^{(-1/\varphi)}$	0.7427	Fixed point of $x \to \varphi^{(-x)}$
3	$r_c = 1 - 1/\varphi^4$	0.8541	Band boundary count (4 boundaries in 5-band partition)
4	cos(1/φ)	0.8150	Spectral phase at AAH frequency
5	σ_3 (matter core)	0.0728	Eigenvalue position of center band / half-range
6	σ_4 (entropy wall)	0.5594	Outer gap edge position / half-range
7	σ_2, σ_wall	0.2350, 0.3972	Inner gap edge, wall center
8	W (gap fraction)	0.4671	$W = (2 + \varphi^{(1/\varphi^2)})/\varphi^4$ — the W theorem
9	N = 294	294	Spectral topology: 233+55+5+1 (Fibonacci Zeckendorf)
10	$\gamma_{dc} = 4$	4	Band boundaries / Chern number sum (topological)
11	δ_k (metallic means)	various	$\delta_k = (k + \sqrt{k^2+4})/2$; selects crystal symmetry
12	F(n) (Fibonacci)	sequence	Gap counts at each recursion level

Step 3: The Cantor Node Architecture

Any physical structure at characteristic radius R (an atom, a star, the observable universe) has five layers at fixed dimensionless positions: r_core = R × 0.0728 (σ_3, matter core), r_inner = R × 0.2350 (σ_2, inner wall), r_photo = R × 0.3672 (cos(1/φ), photosphere), r_shell = R × 0.3972 (σ_wall, density peak), r_outer = R × 0.5594 (σ_4, entropy maximum). These five ratios are extracted once from the eigensolver and applied identically at every scale. The recursion is

self-similar: the center band σ_3 contains sub-gaps that reproduce the same five-sector architecture at reduced scale.

Step 4: The Universal Observable Formula

To predict any physical observable O, multiply the appropriate energy or length scale U at the observable's bracket by a product of the twelve primitives:

$$O = U \times W^k \times (\sigma_i \text{ or } r_c \text{ or } H)^m \times \varphi^p \times \delta_n^q \times F(r) \times (\text{integer or } \pi \text{ or cos})$$

where U is: J = 10.578 eV (hopping integral) for stellar/thermal phenomena; Ry = 13.606 eV (Rydberg) for atomic/chemical; λ_C (Compton wavelength) for nuclear; a_0 (Bohr radius) for molecular geometry; l_P (Planck length) for cosmological. The exponents k, m, p, q, r are small integers determined by the physics of the observable (gap labeling, Chern count, bracket depth, metallic mean symmetry). They are not fitted.

Step 5: The W Theorem

The universal gap fraction W satisfies the exact identity $W \times \varphi^4 = 2 + \varphi^{1/\varphi^2}$, verified to machine precision (error: 2.22×10^{-16}). Every component traces to the axiom: $\varphi^4 = (\varphi^2 = \varphi+1)^2$; the exponent $1/\varphi^2 = \varphi-1$ from the axiom rearranged; 2 from the boundary law endpoint count. The gap fraction is not a fitted parameter—it is the axiom evaluating itself.

The spectral connection to the DGY (2016) thermodynamic pressure function P(t) is established through the five-sector decomposition: weighting the Cantor sectors by the boundary law and hinge fixed point produces W exactly. The mean-field check $W \approx 2 \cdot d_H \cdot (1-d_H)$, where $d_H \approx 0.3725$ is the Hausdorff dimension from Bowen's equation $P(d_H) = 0$, holds to 0.077%. The small discrepancy arises because the true Cantor construction uses the five-sector hierarchy rather than binary splitting.

Computational Tools and LLM Disclosure

All eigenvalue computations use Python 3 with NumPy and SciPy (standard eigensolver). The spectral constants were independently verified at Fibonacci lattice sizes D = 55 through D = 1597. Formula discovery was assisted by two large language models: Claude (Anthropic) for iterative computation, document preparation, and adversarial testing; and Grok (xAI) for independent verification, literature search across all 49 papers citing DGY (2016), and confirmation that no prior work extracts coupling constants from the Fibonacci gap structure. All LLM-assisted computations were verified by direct numerical evaluation. The W theorem was discovered during a Claude session on March 21, 2026; the five-sector P(t) connection was verified by Grok the same day. In accordance with Research Square policy, LLMs are not listed as authors, as they cannot bear accountability for the work.

Validation Results

We present validation against independently measured quantities organized by domain. All predictions use zero adjustable parameters. Errors are calculated as |predicted − observed|/observed × 100%.

Selected Precision Benchmarks

Observable	Predicted	Observed	Error	Domain
Entropy max. position (σ_4)	1.408380 a_0	1.408377 a_0	0.0002 1%	Atomic
232 attoseconds	(D−1)×1 as	232.0 as	0.005%	Nuclear
$\sin^2\theta$_W (Weinberg)	σ_3×σ_wall×F(6)	0.23122	0.026%	Particle
Magic angle (graphene)	α_{53}	1.08°	0.06%	Cond. matter
Proton charge radius	0.8426 fm	0.8414 fm	0.14%	Nuclear
Fine structure α^{-1}	N×W = 137.34	137.036	0.22%	QED
M_W/m_p	$\varphi^2 \times W^{-2} \times \delta_7$	85.657	0.002%	Particle
M_H/m_p	$\varphi^2 \times \delta_7^2$	133.49	0.015%	Particle
Si band gap	$\sigma_2^2 \times 1/\delta_4 \times \delta_8$	1.12 eV	0.005%	Semiconductor
Au work function	$H \times 4 \times 1/\delta_6$	5.1 eV	0.002%	Materials
N-SmA transition α(r)	$(2/3)((r-r_c)/(1-r_c))^4$	11 compounds	RMS 0.033	Cond. matter
NGC 3198 v_flat	149 km/s	148 km/s	1%	Galaxy
Ω_DE (dark energy)	$W^2+W = 0.6853$	0.685	0.05%	Cosmology
Gravity/EM hierarchy	$(\sqrt{(1-W^2)}/\varphi)^{136}$	10^{-36}	1.1% log	Gravity
Atomic radii (54 elements)	$\sqrt{(1+(\Theta \times BOS)^2)}$	54 elements	6.2% mean	Chemistry

The method produces predictions spanning 15 independent physical domains—nuclear physics, atomic structure, chemistry, stellar astrophysics, condensed matter, particle physics, semiconductor physics, materials science, cosmology, galaxy dynamics, quantum Hall physics, liquid crystals, biology (microtubule gating), gravitational physics, and exoplanet science—from the same twelve primitives extracted from one 233×233 matrix. We are aware of no other zero-parameter method with comparable scope.

Honest Assessment of Limitations

We emphasize several limitations. (1) The extraction of W from the five-sector pressure decomposition, while verified numerically, does not yet constitute a formal mathematical proof in the sense of DGY's theorems. (2) The exponents k, m, p, q in the universal formula are determined by physical reasoning (gap labeling, symmetry matching), not by a single closed algorithm—an element of judgment remains. (3) The 54-element atomic radius predictions have a mean error of 6.2%, which is useful but not spectroscopic precision. (4) The galaxy rotation results, while competitive with NFW and MOND, have been adversarially tested (4 rounds, Grok/xAI) and contain identified open questions (gravity × gate compensation is emergent, not proven as a theorem). (5) The W–d_H quadratic connection (0.077%) has not been proven

exact. We report these gaps because transparency strengthens, rather than weakens, the case for the method.

Discussion

The central claim of this article is methodological, not theoretical: given the critical Fibonacci spectrum, one can extract a small set of dimensionless primitives that reproduce physical observables across many domains at sub-percent accuracy. Whether this reflects a deep truth about the structure of the vacuum (as one of us believes) or an unexplained but fertile pattern (as a skeptical reader may prefer) is a question we leave open. The predictions are falsifiable, the computation is reproducible, and the code is publicly available.

The W theorem—an exact algebraic-transcendental identity connecting the gap fraction to the golden ratio through self-referential exponentiation—appears to be new. We have not found it in the spectral theory literature (confirmed by exhaustive search of all papers citing DGY 2016). If it is already known under a different name, we would be grateful to be informed.

Li and Boyle's (2023) result that Fibonacci tilings are quantum error-correcting codes suggests that the spectral constants extracted here are protected against local perturbations. This may explain the robustness of the predictions: the gap structure that determines $\alpha^{-1} = N \times W$ is topologically protected, not fragile.

We offer this method in the spirit of Leonardo's *Liber Abaci*: not as a finished theory, but as a practical tool that works surprisingly well and invites explanation. If the Fibonacci Hamiltonian at $V = 2J$ truly encodes the coupling constants of the universe, then the mathematicians in our table above built the foundation decades before anyone thought to look for physics in it. We are merely the first to try.

Data Availability

All code, eigenvalue data, and verification scripts are publicly available at: https://github.com/thusmann5327/Unified_Theory_Physics. The spectral constants can be reproduced by any standard eigensolver (NumPy, MATLAB, Mathematica) applied to the 233×233 AAH Hamiltonian at $V = 2J$, $\alpha = 1/\varphi$. No proprietary data or software is required.

Competing Interests

T.A.H. holds 16 provisional patent applications (63/995,401 through 63/998,394) related to engineering applications of the framework. The spectral constants, mathematical identities, and physical predictions described in this article are not subject to patent claims and are freely available for academic use under CC BY 4.0.

Acknowledgments

We owe an enormous debt to the spectral theorists who built the rigorous foundation. The complete analysis of Damanik, Gorodetski, and Yessen (2016) provided the exact

thermodynamic pressure P(t) and dimension formula that allowed us to extract the universal gap fraction W. Maciá's (2017) two-cluster decomposition directly revealed the five-sector Cantor structure used throughout the framework. Jagannathan's (2021) review clarified the hidden-dimensional embedding and Chern-number gap labeling that connect the one-dimensional lattice to three-dimensional gravity. Li and Boyle's (2023) discovery that Fibonacci tilings are quantum error-correcting codes supplied the information-theoretic protection that makes the constants robust. The trace-map techniques pioneered by Kohmoto, Sutherland, and Tang (1983) and Sütő (1987), together with Bowen's thermodynamic formalism, closed the loop from the pure axiom $\varphi^2 = \varphi + 1$ to the physical constants and emergent gravity. Without these works, this synthesis would not exist.

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